

DEVA ANAND

devaanand@umass.edu • (413) 315-1715 • devaanand.com • linkedin.com/in/devaaa • github.com/Deva-1903

Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR text-analysis platform (Python, **FastAPI**, sentence-transformers, React) end to end with full technical ownership delegated by the PI, translating an ambiguous brief into a working self-serve web tool with no supporting engineering team, self-teaching and "vibe-coding" fast on a **Claude Code** multi-agent workflow.
- Built the platform's versioned data backbone (YAML model registry as single source of truth, a **99-construct** library from **38 questionnaires**, structured data-quality warnings, DB-persisted job queue), grew the backend to **40 hermetic CI tests**, and cost-engineered the deploy plan to a **\$0–60/yr** budget.

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building AI features into an operational decision-support workflow: a provider-facing app (Next.js + Firebase/GCP) that surfaces covered, guideline-backed medication alternatives inside the EHR prescribing flow.
- Extending a **Firestore Cloud Functions** backend that builds recommendations from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility (auditable, grounded AI-generated insights).

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly submissions** via Mayden iaptus **REST integrations** and aggregation queries; contributed to **\$340K+** in revenue.
- Architected a full-stack internal annotation tool (React, Node.js, MongoDB) used daily by the AI team, replacing spreadsheet workflows with role-based REST APIs and cutting manual operations by **100+ hours/month** while lifting labeling throughput **60%**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service (Python/FastAPI) that turns open-ended queries into structured entity tables via typed retrieval planning, deterministic extraction with LLM fallback, and **cell-level provenance** for every value; hardened with **150+ tests** and provider routing across OpenAI/Groq.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Optimized a Python/PySpark ETL + analytics pipeline over a **~1.4B-row** dataset with staged schema-aware reads, column pruning, broadcast joins, AQE, and skew-join handling, cutting end-to-end runtime **~25%** and profiling stage timings via Spark UI.

align-ops – LLM Serving & Post-Training Ops PyTorch, TRL, vLLM, FastAPI, Redis, Docker

- Built a FastAPI gateway in front of a vLLM server adding **Redis** response caching and async micro-batching, cutting repeated-request latency **2.1s → 24ms** and lifting throughput **1.9x**; containerized with Docker and instrumented with Prometheus/Grafana dashboards.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Bash

AI & LLM: OpenAI API, LLM agents, RAG, prompt engineering, sentence-transformers, PyTorch, evaluation

Backend & Web: FastAPI, Node.js, Express, React, Next.js, REST APIs, JSON, async services

Data: pandas, NumPy, PySpark, ETL, MongoDB, PostgreSQL, SQLite, Redis

AI Dev & Cloud: Claude Code, GitHub Copilot, OpenAI Codex, Cursor, Docker, Git, AWS, GCP, Azure

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN operations platform with Stripe payments and cron automation; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, ProjectsHut. Tutored **30+ students** in programming at Sayur (non-profit, Tamil Nadu).